Bell States

Lets Recap..

- Why do we need Quantum Cryptography?
- Qubits
- Superposition
- Measurement
- Entanglement
- **Bell states**

Bell States Equations

- Bell states form orthonormal basis in two qubit system.
- These are four maximum **Entangled** states.

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

But we already had standard(Computational) basis in 2 qubit system, then Why Bell basis?

$$|AX\rangle = |A\rangle \otimes |X\rangle = \begin{bmatrix}1\\0\end{bmatrix} \otimes \begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

$$|AY\rangle = |A\rangle \otimes |Y\rangle = \begin{bmatrix}1\\0\end{bmatrix} \otimes \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}0\\1\\0\\0\end{bmatrix}$$

$$|BX\rangle = |B\rangle \otimes |X\rangle = \begin{bmatrix}0\\1\end{bmatrix} \otimes \begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}0\\0\\1\\0\end{bmatrix}$$

$$|BY\rangle = |B\rangle \otimes |Y\rangle = \begin{bmatrix}0\\1\end{bmatrix} \otimes \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\0\\1\end{bmatrix}$$

Answer: Because Bell basis helps to reveal entanglement

In the computational basis:

- Entanglement is hidden inside coefficients
- Hard to see correlations directly

In the Bell basis:

- Each basis vector is already maximally entangled
- Entanglement becomes explicit
- Correlations become simple

Example

For,

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Measuring in computational basis gives:

- 50% → 00
- 50% → 11

But from this you **cannot tell** whether the state was:

- Φ^+
- Φ^-

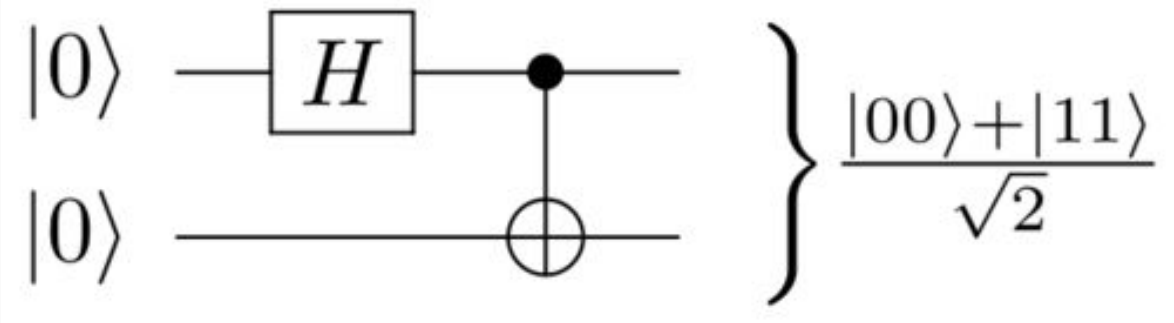
➢ **Information about phase is lost.**

Bell basis measurement preserves it.

How these states are formed?

Lets Derive this state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



Step 1: Start with Initial State

$$|00\rangle$$

Step 2: Apply Hadamard to First Qubit

Hadamard creates superposition:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Apply to first qubit:

$$|00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

➢ Now system is in superposition but **not yet entangled**.

Step 3: Apply CNOT Gate

CNOT rule:

- If control qubit = 1 → flip target
- If control qubit = 0 → do nothing

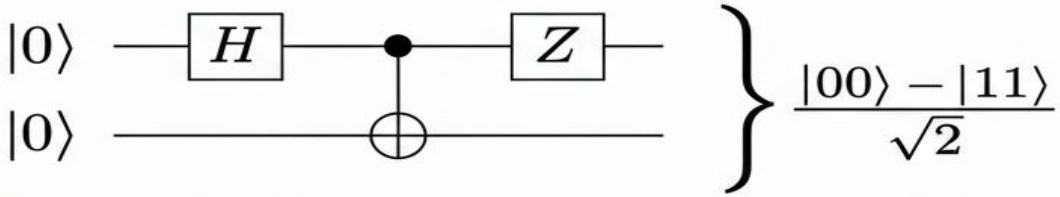
Step 4: Final State

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

How these states are formed?

Lets Derive this state:

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$



Step 1: Start with Initial State

$$|00\rangle$$

Step 2: Apply Hadamard to First Qubit

Hadamard creates superposition:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

New state: $\frac{|00\rangle + |10\rangle}{\sqrt{2}}$

Step 3: Apply CNOT Gate

CNOT rule:

- If control qubit = 1 → flip target
- If control qubit = 0 → do nothing

Applying this logic, the state becomes: $\frac{|00\rangle + |11\rangle}{\sqrt{2}}$

Step 4: Apply Z Gate to First Qubit

The Z gate applies to the first qubit of the state.
Applying this logic to the previous state gives the final state:
Z-gate rule: $Z|0\rangle = |0\rangle, Z|1\rangle = -|1\rangle$

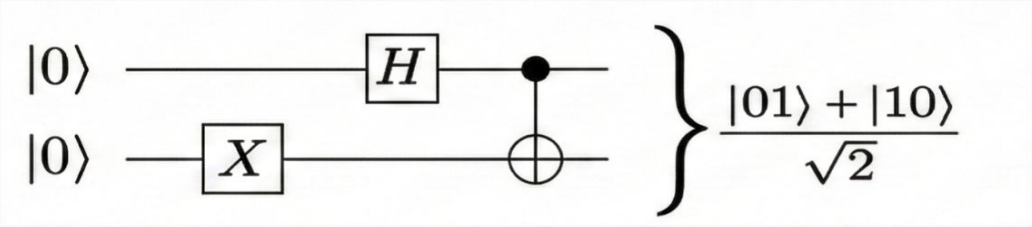
Step 5: Final State

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

How these states are formed?

Lets Derive this state:

$$|\Psi^+ = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$



Step 1: Start with Initial State

$|00\rangle$

Step 2: Apply X Gate to 2nd Qubit

Apply X gate to the second qubit.
The NOT gate operation is $X|0\rangle = |1\rangle$.

Step 3: Apply Hadamard Gate on first qubit

The Hadamard gate puts the first qubit into a superposition.

Mathematical rule: $H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$.

CNOT logic:

- If control qubit = 1 → flip target
- If control qubit = 0 → do nothing

Applying this logic, the |01> term remains (control 0) and the |11> term becomes |10> (control 1).

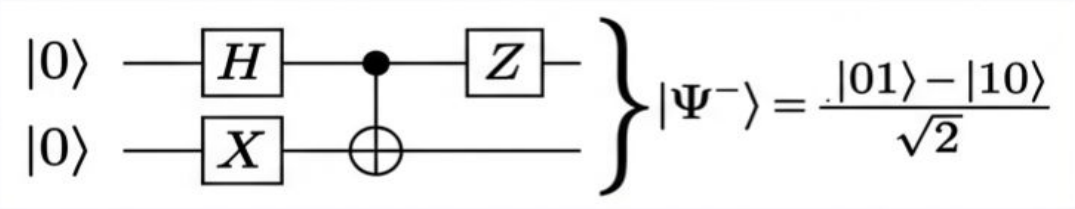
Step 5: Final State

$$|\Psi^+ = \frac{|01\rangle + |10\rangle}{\sqrt{2}}.$$

How these states are formed?

Lets Derive this state:

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$



Step 1: Start with Initial State

Step 2: Apply X Gate on Second Qubit

Apply X Gate on Second Qubit. X flips the qubit. The NOT gate operation is $X|0\rangle = |1\rangle$.

Step 3: Apply Hadamard Gate on First Qubit

Apply Hadamard Gate on First Qubit. Hadamard creates superposition:

Hadamard rule: $H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$.

Step 4: Apply CNOT Gate

CNOT rule:

- If control qubit = 1 → flip target
- If control qubit = 0 → do nothing

Applying this logic, the state becomes: $\frac{|01\rangle + |10\rangle}{\sqrt{2}}$.

Step 5: Apply Z Gate on First Qubit

Apply Z Gate on First Qubit. Z gate rule:

- $Z|0\rangle = |0\rangle$
- $Z|1\rangle = -|1\rangle$

Applying this rule to the previous state, we arrive at the Final State: $|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$.

Uses of Bell States

- Teleportation
- E91 protocol
- Superdense Coding
- Entanglement Swapping
- Bell Inequality Tests
- Quantum Error Correction

EPR Paradox and Local Hidden Variables

- Proposed by Einstein, Podolsky, and Rosen (EPR) to question the completeness of quantum mechanics.

- **Locality**: The notion that no influence can propagate faster than light.
- **Realism**: The measurement outcomes are determined before the measurement.

- Einstein famously derided quantum state collapse as ”**spooky action-at-a-distance**”.

This paradox was questioning the completeness of Quantum mechanics which remained unsolved for many decades.

Bell's Inequality

- John Stewart Bell proved that LHV theories and quantum mechanics make different statistical predictions.
- He derived a mathematical framework to test these differences.
- **The Core Result:** If an experiment violates a "Bell inequality," then nature cannot be described by any local hidden variable theory

CHSH Inequality

Why Prove the CHSH Inequality?

- 1935: EPR Paradox (Einstein, Podolsky, Rosen)
- They claimed quantum mechanics is **incomplete**
- Proposed **local hidden variables** (secret instructions)
- Two key ideas:
  - **Realism**: Outcomes are predetermined
  - **Locality**: No influence faster than light

Goal: A mathematical test to decide: Einstein or QM?

Bell (1964) and CHSH gave us the test

What Does S Mean?

- S is a number measuring **total correlation strength**
- Measures how Alice and Bob's results are linked across 4 setting combinations

The Correlation Score:

- **Classical world**: max score $= 2$
- **Quantum world**: can reach 2.828

Proves something impossible for classical physics!

The Formula of S

Alice and Bob share the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\Big(|00\rangle + |11\rangle\Big)$$

Quantum Correlation $E(A, B) = P_{00} - P_{01} - P_{10} + P_{11}$

CHSH Quantity S:

$$S = E(A, B) + E(A, B') + E(A', B) - E(A', B')$$

Alice's Measurement Bases

- **A** (Z-basis): $\{|0\rangle, |1\rangle\}$
- **A'** (X-basis): $\{|+\rangle, |-\rangle\}$

Circuit Strategy:

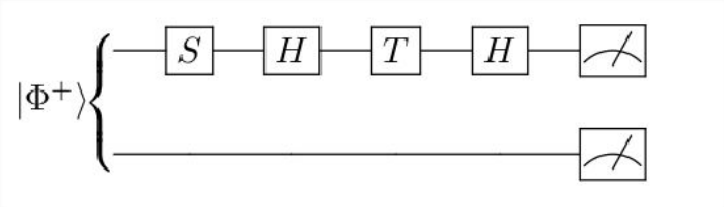
- A $\rightarrow$ Direct Z-measurement
- A' $\rightarrow$ Apply **H gate** then Z-measurement

Bob's Special Bases B and B'

$$B = \left\{ \frac{(1+\sqrt{2})|0\rangle + |1\rangle}{\sqrt{4+2\sqrt{2}}}, \frac{(1-\sqrt{2})|0\rangle + |1\rangle}{\sqrt{4-2\sqrt{2}}} \right\} \quad B' = \left\{ \frac{(-1-\sqrt{2})|0\rangle + |1\rangle}{\sqrt{4+2\sqrt{2}}}, \frac{(-1+\sqrt{2})|0\rangle + |1\rangle}{\sqrt{4-2\sqrt{2}}} \right\}$$

Circuit Implementation:

- Apply **S, H, T, H** gates $\rightarrow$ then Z-measurement
- B' uses **S, H, T†, H** gates



Step-by-Step Quantum Proof

Final state for AB before measurement:

$$\frac{1+e^{i\pi/4}}{2\sqrt{2}}|00\rangle + \frac{1-e^{i\pi/4}}{2\sqrt{2}}|01\rangle + i\frac{1-e^{i\pi/4}}{2\sqrt{2}}|10\rangle + i\frac{1+e^{i\pi/4}}{2\sqrt{2}}|11\rangle$$

Probabilities:

$$P_{00} = \tfrac{2+\sqrt{2}}{8}, \quad P_{01} = \tfrac{2-\sqrt{2}}{8}, \quad P_{10} = \tfrac{2-\sqrt{2}}{8}, \quad P_{11} = \tfrac{2+\sqrt{2}}{8}$$

Correlation:

$$E(A,B) = \frac{1}{\sqrt{2}} \approx 0.707$$

All Four Quantum Correlations

- $E(A, B) = +1/\sqrt{2}$
- $E(A, B') = +1/\sqrt{2}$
- $E(A', B) = +1/\sqrt{2}$
- $E(A', B') = -1/\sqrt{2}$

The negative sign is the key to the violation!

Quantum Prediction for S

$$S = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right)$$

$$S = \frac{4}{\sqrt{2}} = 2\sqrt{2} \approx 2.828$$

Tsirelson's Bound: $|S| \leq 2\sqrt{2}$

The Violation

- Classical limit: $|S| \leq 2$
- Quantum Prediction: $S \approx 2.828$

$$\mathbf{2.828 > 2}$$

- Local hidden variables are **wrong**.
- Entanglement is a real, non-classical phenomenon.

References

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5. https://medium.com/into-quantum/what-are-bell-states-how-to-express-quantum-systems-using-them-1b6825701332

Thank you